



DAVID ÁLVAREZ ROSA

Mathematician · Engineer · Dublin



March 2022

Low-latency engineer in high-frequency options trading at Susquehanna. Represents Ireland on the ISO C++ standards committee. Previously at Amazon. Holds a BSc in Mathematics, a BEng in Industrial Engineering, and an MSc in Artificial Intelligence.

EXPERIENCE

Software Engineer *Jul 2024–Present*
Susquehanna International Group · Dublin, Ireland

High-frequency options trading. Low-latency market data and trading signals. Mentor and interviewer.

Member *Jul 2026–Present*
ISO C++ Committee (WG21) · Ireland

National expert. Representing Ireland.

Software Engineer II *Mar 2022–Aug 2024*
Amazon · Madrid, Spain

Designed systems for 10M+ monthly active customers. Contributed to 100+ internal repos. Won org hackathon. Promoted in 18 months (top 5%). Mentored 3. On-call. Interviewer.

Machine Learning Engineer *Apr 2024–Jul 2024*
Sopra Steria · Remote

Researched, designed, and built a semantic cache for LLMs.

Risk Analyst *Sep 2021–Mar 2022*
Deloitte · Madrid, Spain

Quantitative analysis of cybersecurity risk for banks.

Visiting Researcher *Sep 2020–Jun 2021*
Vector Institute · University of Toronto, Canada

Thesis on multimodal learning (10/10, A+) *recomprehension.com*.

Machine Learning Engineer *Sep 2019–Feb 2020*
BCN eMotorsport · Barcelona, Spain

LiDAR lead and computer vision for a fully autonomous car.

EDUCATION

MSc in Artificial Intelligence *9.00/10*
International University of La Rioja

PhD-qualifying research program. Honors in 6 subjects.

MSc in Mathematics
National University of Distance Education

Math-lover part-time student. Dropout (joined Amazon).

BSc in Mathematics *8.12/10 (top 10%)*
Polytechnic University of Catalonia (CFIS)

Rigorous, proof-oriented program. Honors in 9 subjects.

BEng in Industrial Engineering *8.03/10 (top 2%)*
Polytechnic University of Catalonia (CFIS)

Multidisciplinary engineering program. Honors in 14 subjects.

SKILLS

C++ · Python · Linux · Performance

CERTIFICATIONS

Advanced English (C1) *Cambridge*

Machine Learning *Stanford University*

Deep Learning *deeplearning.ai*

Blockchain *Hong Kong University*

Nova Talent Member *Nova*

HONORS & AWARDS

Mathematical Olympiad
Silver (local), honors (national).

Physics Olympiad
Gold (local), silver (national).

Mobility Scholarship
Research thesis at Toronto (€6k).

Tuition and Housing Scholarship
University tuition and housing (€19k).

General Scholarship
Full tuition plus annual stipend (€11k).

VOLUNTEERING

Mathematics Tutor
Training for the Mathematical Olympiads.

Food Bank Volunteer
Food collection at Banco de Alimentos.

LANGUAGES

English *Proficient*

Spanish *Native*

Catalan *Intermediate*

CONTACT

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